

PART-B

Name of the Student: Jain Mayank Gautam bhai
Signature of the Student: [Signature] Roll No.: 230123026
Signature of the Invigilator: [Signature]

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	7	5	11	23

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX} \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

We know $R^2 = 1 - \frac{\text{PRESS}}{S_{yy}}$

$$= 1 - \frac{\sum_{i=1}^n (e_i)^2}{\sum_{i=1}^n (H_{ii})^2} \times \frac{1}{S_{yy}} = 1 - \frac{\sum_{i=1}^n e_i^2}{S_{yy}}$$

$$Y = [y]$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_p \end{bmatrix}$$

Also

$$r_1^2 =$$

$$S_{yy}^{-1/2} S_{yy}^{-1} S_{yx}^{-1} S_{yx} S_{yy}^{-1/2}$$

because as $Y = 1 \times 1$ so there will only 1 canonical correlation which is given by above.

$$H_2 = X(X^T X)^{-1} X^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \left(\frac{1}{x_1^2 + x_2^2 + \dots + x_n^2} \right) \alpha \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$$

$$= \frac{1}{x_1^2 + x_2^2 + \dots + x_n^2} \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_n \\ x_2 x_1 & x_2^2 & \dots & x_2 x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n x_1 & x_n x_2 & \dots & x_n^2 \end{bmatrix}$$

$$\Rightarrow H_{11} = \frac{\alpha^2}{\sum_{j=1}^n x_j^2}$$

$$R^2 = \left[\frac{1}{\sum_{j=1}^n y_j^2} \right] \begin{bmatrix} y_1 x_1 & y_1 x_2 & \dots & y_1 x_n \\ y_2 x_1 & y_2 x_2 & \dots & y_2 x_n \\ \vdots & \vdots & \ddots & \vdots \\ y_n x_1 & y_n x_2 & \dots & y_n x_n \end{bmatrix} \begin{bmatrix} \frac{\alpha^2}{\sum_{j=1}^n x_j^2} & \frac{\alpha^2 x_1 x_2}{\sum_{j=1}^n x_j^2} & \dots & \frac{\alpha^2 x_1 x_n}{\sum_{j=1}^n x_j^2} \\ \frac{\alpha^2 x_2 x_1}{\sum_{j=1}^n x_j^2} & \frac{\alpha^2 x_2^2}{\sum_{j=1}^n x_j^2} & \dots & \frac{\alpha^2 x_2 x_n}{\sum_{j=1}^n x_j^2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\alpha^2 x_n x_1}{\sum_{j=1}^n x_j^2} & \frac{\alpha^2 x_n x_2}{\sum_{j=1}^n x_j^2} & \dots & \frac{\alpha^2 x_n^2}{\sum_{j=1}^n x_j^2} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_T} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

$$R^2 = S_{yy}^{-1/2} S_{yx}^T S_{xx}^{-1} S_{yx} S_{yy}^{1/2}$$

$$= \left(\frac{1}{\sum (y_i - \bar{y})^2} \right) S_{yx}^T S_{xx}^{-1} S_{yx} \left(\frac{1}{\sum (y_i - \bar{y})^2} \right)$$

$$S_{yx}^T = \begin{bmatrix} \sum (y_i - \bar{y})(x_{i1} - \bar{x}_1) & \sum (y_i - \bar{y})(x_{i2} - \bar{x}_2) & \dots & \sum (y_i - \bar{y})(x_{in} - \bar{x}_n) \end{bmatrix}$$

$$S_{yx} = \begin{bmatrix} \sum (y_i - \bar{y})(x_{i1} - \bar{x}_1) \\ \sum (y_i - \bar{y})(x_{i2} - \bar{x}_2) \\ \vdots \\ \sum (y_i - \bar{y})(x_{in} - \bar{x}_n) \end{bmatrix}$$

$$S_{xx} = \begin{bmatrix} \sum (x_{i1} - \bar{x}_1)^2 & \sum (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) & \dots & \sum (x_{i1} - \bar{x}_1)(x_{in} - \bar{x}_n) \\ \sum (x_{i2} - \bar{x}_2)(x_{i1} - \bar{x}_1) & \sum (x_{i2} - \bar{x}_2)^2 & \dots & \sum (x_{i2} - \bar{x}_2)(x_{in} - \bar{x}_n) \\ \vdots & \vdots & \ddots & \vdots \\ \sum (x_{in} - \bar{x}_n)(x_{i1} - \bar{x}_1) & \sum (x_{in} - \bar{x}_n)(x_{i2} - \bar{x}_2) & \dots & \sum (x_{in} - \bar{x}_n)^2 \end{bmatrix}$$

00

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

(a) we know for m factors, $\mathbf{e}_i \rightarrow$ are column vectors

$$\tilde{\mathbf{L}} = [\sqrt{d_1} \mathbf{e}_1, \sqrt{d_2} \mathbf{e}_2, \dots, \sqrt{d_m} \mathbf{e}_m]$$

where $d_1 \geq d_2 \geq \dots \geq d_m \geq \dots \geq d_p$

so here largest eigen value is $d_3 = 4.2$

2nd largest is $d_2 = 2.1$, $\tilde{\mathbf{L}} = [\sqrt{d_3} \mathbf{e}_3 \mid \sqrt{d_2} \mathbf{e}_2]$

$$\Rightarrow \tilde{\mathbf{L}} = \left[\begin{array}{c|c} \sqrt{4.2} \mathbf{e}_3 & \sqrt{2.1} \mathbf{e}_2 \end{array} \right]_{4 \times 2}$$

$\uparrow$
column 1
 $\uparrow$
column 2

02

(b) we know

$$\tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2$$

$$= (0.8)^2 + (-0.4)^2$$

$$= 0.8$$

01

$$(c) \quad \tilde{L} = \begin{bmatrix} \sqrt{4.2} e_{31} & \sqrt{2.1} e_{21} \\ \sqrt{4.2} e_{32} & \sqrt{2.1} e_{22} \\ \sqrt{4.2} e_{33} & \sqrt{2.1} e_{23} \\ \sqrt{4.2} e_{34} & \sqrt{2.1} e_{24} \end{bmatrix}$$

$$\sum_{i=1}^4 \tilde{h}_i^2 = (4.2) e_{31}^2 + (2.1) e_{21}^2 + 4.2 e_{32}^2 + 2.1 e_{22}^2 + 4.2 e_{33}^2 + 2.1 e_{23}^2 + 4.2 e_{34}^2 + 2.1 e_{24}^2$$

$$= (4.2) (e_{31}^2 + e_{32}^2 + e_{33}^2 + e_{34}^2) + (2.1) (e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2)$$

$$= 4.2 + 2.1 \quad (\because e_3 \text{ \& } e_2 \text{ are normalized eigen vectors} \Rightarrow \sum_{i=1}^4 e_i^2 = 1)$$

04

= sum of largest two eigen values of S.

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

~~Ans~~ As $\frac{\bar{X}_n - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \sim N(0, 1)$

By delta method.
Asymptotic ~~distribution~~ ~~variance~~ of $g(\bar{X}_n)$ will be.

$$g(\bar{X}_n) - g(\mu) \approx g'(\mu) N\left(0, \frac{\mu^2}{n}\right)$$

Variance of $g(\bar{X}_n) = g'^2(\mu) \times \frac{\mu^2}{n}$

For $g'^2(\mu) \times \frac{\mu^2}{n}$ independent of n

Say $\frac{1}{n}$
 $\Rightarrow g'^2(\mu) \frac{\mu^2}{n} = \frac{1}{n}$

$$\Rightarrow g'(\mu) = \frac{1}{\mu^2}$$

$$\Rightarrow g'(\mu) = \frac{1}{\mu}$$

05 $\Rightarrow g(\mu) = \log(\mu)$ - take,

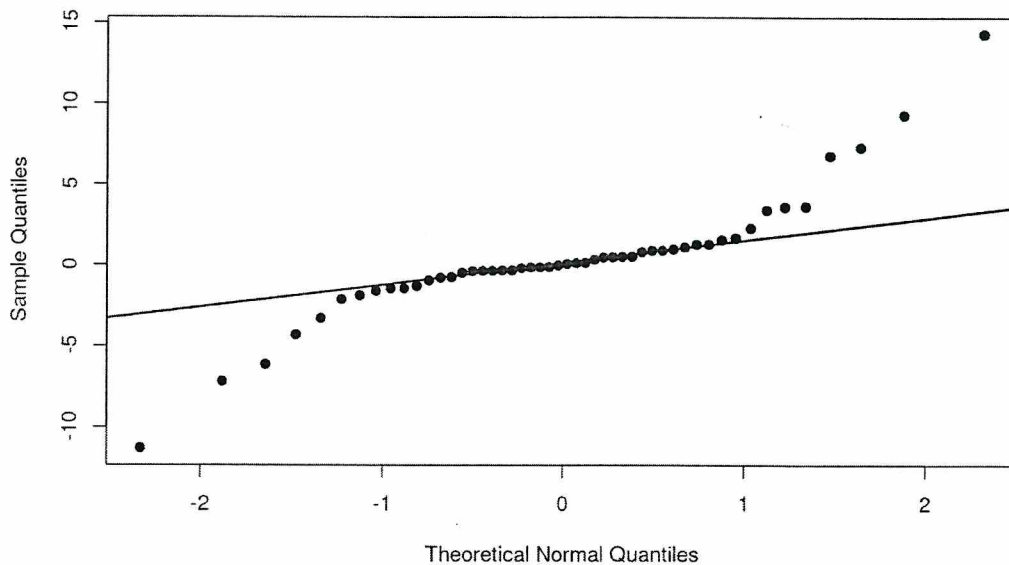
so if we take $g(\mu) = \log \mu$ then asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Ans, $\log(\bar{X}_n) - \log(\mu) \approx \frac{1}{\mu} N\left(0, \frac{\mu^2}{n}\right) \approx N\left(0, \frac{1}{n}\right)$

$$\Rightarrow \log(\bar{X}_n) \approx N\left(\log(\mu), \frac{1}{n}\right)$$

$$\text{Var}(\log(\bar{X}_n)) \approx \frac{1}{n} \quad (\text{independent of } \mu).$$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



As Quantile-Quantile plot is not giving $y=x$ so normality condition is violated, but as it follows straight line with small deviation ~~but~~ and somewhat flatter ~~so~~ it shows short tail deviation and some transformation can bring it in normal form.

so distribution can be log-normal instead of normal ~~so~~.

01

01

ADDITIONAL PAGES FOR ANSWERS
